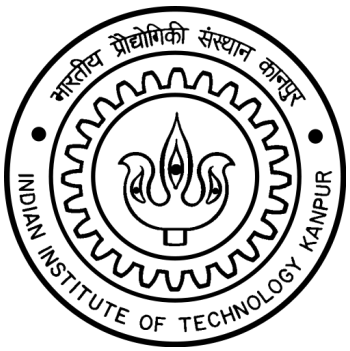


SPA617

RADIO ASTRONOMY LABORATORY

Antenna Characterization

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1. Introduction

1.1. Radio Waves: What I Knew Before the Experiment

The primary knowledge about radio waves that I had before this laboratory was limited mainly to their frequency and wavelength, and their position within the electromagnetic spectrum. In the context of physics, I had encountered radio waves primarily as electromagnetic radiation and had not explored their applications in atomic/molecular spectroscopy or astronomy in much detail.

I was, however, aware that unlike many other regions of the electromagnetic spectrum, a significant portion of the radio spectrum is transmitted through the atmosphere, although the transparency depends on the frequency and atmospheric conditions. This makes ground-based radio observations possible over a broad range of wavelengths for radio waves.

Beyond these ideas, my understanding of radio waves was largely limited to their basic electromagnetic properties. I had not previously considered in detail how radio-frequency radiation could be detected using antennas, how the received signal depends on the incident direction, or how antenna properties determine the ability to detect radiation from different directions.

1.2. Antennas: What I Knew Before the Experiment

I had always been aware of the importance of antennas, as they have been a part of numerous technologies encountered in everyday life. I was familiar with the long monopole antennas used in cars, radios, and other communication systems, as well as dish antennas commonly used in televisions. I had also encountered antennas in remote-controlled vehicles and drones, where they are used for wireless communication and control. Thus, while the importance and widespread use of antennas were familiar to me, I had not previously considered in detail how an antenna actually transmits or receives electromagnetic radiation.

During my undergraduate studies, I had also encountered the basic principles of LiDAR, which provided me with some familiarity with the use of electromagnetic radiation for remote sensing. However, my understanding of antennas themselves remained largely qualitative. The laboratory provided a more detailed introduction to the physical principles governing antenna operation, including their directional response and dependence on orientation and polarization. The dependence of length on the resonance frequency of the antennas was also something that I clearly understood in the laboratory. This was particularly interesting to me, as it connected the familiar everyday presence of antennas with the underlying electromagnetic physics.

1.3. What the Experiment Taught Me About Antennas

Several observations and discussions during the experiment helped me develop a more intuitive understanding of an antenna's operation.

- **Construction of an antenna:** Observing the construction of the sensing element provided me with an initial understanding of how the physical dimensions of an antenna influence its resonant frequency.
- **Directional response and directivity:** Before the experiment, I had a general understanding that antennas could transmit and receive radiation over a range of directions. However, this was the first time I experimentally investigated how the response of an antenna varies with direction. By rotating the receiving antenna through different angles and recording the received signal, I was able to observe the directional dependence of the antenna response and identify the direction of maximum response. A discussion with

the instructor also introduced me to how the orientation and directivity of large radio-observational facilities can be characterized measurements involving moving platforms and the motion of astronomical sources caused by the Earth's rotation.

- **Effect of polarization:** The experiment demonstrated the effect of the relative polarization of the transmitting and receiving antennas on the amplitude of the received signal.
- **Interaction with the surroundings:** An important practical observation was the sensitivity of the measured signal to the surroundings of the experiment. The presence of walls, nearby objects, and even people could alter the measured response.

2. Aim

The aim of this experiment is to characterize the radial, directional and polarization-dependent response of a horn antenna using a dipole antenna as the transmitting source.

3. Objectives

The objectives of this experiment are:

- To understand the basic operating principles and construction of dipole and horn antennas.
- To study the variation of the received intensity of the incident radiation of the horn antenna with the distance between the dipole and horn antenna.
- To experimentally determine the directional response of the horn antenna by measuring the received signal as a function of its angular orientation.
- To obtain and compare the antenna response in the E-plane and H-plane configurations.
- To investigate the effect of relative polarization between the transmitting and receiving antennas on the received signal.
- To characterize the beam pattern of the horn antenna and identify features such as the direction of maximum response and beamwidth.
- To understand the effects of reflections, scattering, and other environmental factors on practical antenna measurements.

4. Apparatus

The following apparatus were used for the antenna characterization experiment:

- Horn antenna
- Dipole antenna
- Signal generator
- Spectrum analyzer
- Measuring tape with a least count of 1 mm
- An angular measurement scale

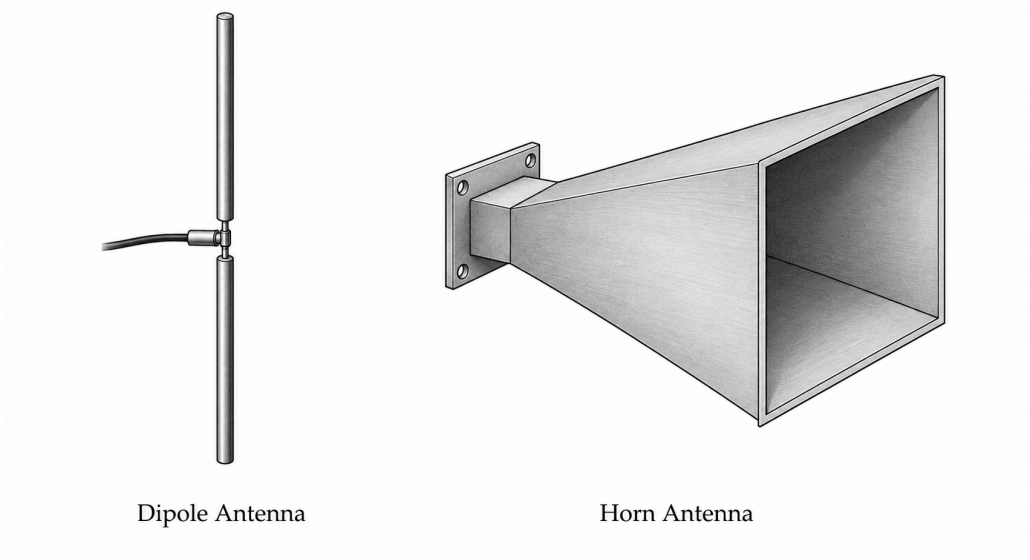


Figure 1: Dipole and horn antennas used in the experiment.

5. Theory useful for Experiment

5.1. Dependence of Received Power on Distance

As an electromagnetic wave propagates through free space, its power density decreases with increasing distance from the transmitting antenna. In the far-field region, the received power approximately follows the inverse-square law,

$$P_r \propto \frac{1}{r^2}, \tag{1}$$

where P_r is the received power and r is the separation between the transmitting and receiving antennas.

5.2. Polarization and Plane Configurations

Polarization describes the orientation of the electric-field vector of an electromagnetic wave. For a linearly polarized wave, the electric field oscillates along a fixed direction perpendicular to the direction of propagation. The receiving antenna responds most strongly when its polarization is aligned with that of the incident wave, while a relative rotation of the polarization axes reduces the strength of the received signal.

In the present experiment, two polarization configurations were considered. In the *co-polarized* configuration, the polarization of the receiving antenna was aligned with that of the transmitted wave. In the *cross-polarized* configuration, the receiving antenna was oriented orthogonally to the co-polarized configuration.

For the antenna, the E-plane and H-plane configurations were defined according to the orientation of the sensing element of the receiving antenna. The E plane was defined to be the plane along the radial direction of the sensing element whereas the H plane was defined to be perpendicular to it. Rotating the antenna in these planes enabled the measurement of the angular variation in the respective planes.

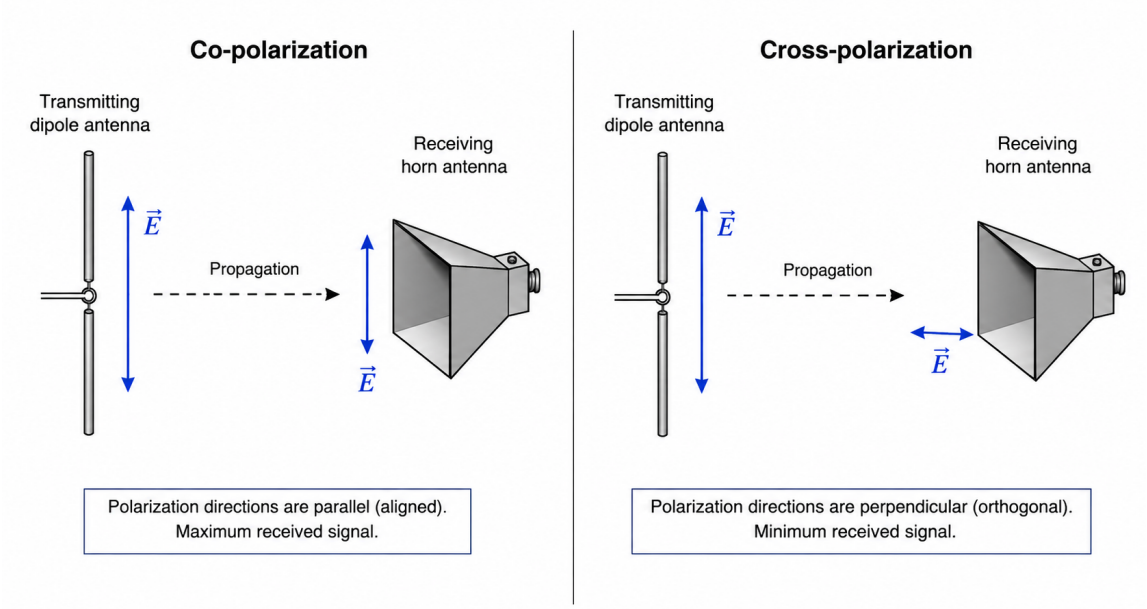


Figure 2: Co-polarization and cross-polarization configurations.

5.3. Directivity

Directivity is a measure of how concentrated the radiation is in the direction of the main lobe compared with an isotropic antenna. It is related to the beam solid angle Ω_A as

$$D = \frac{4\pi}{\Omega_A}. \tag{2}$$

For a practical antenna, the directivity can be estimated from the half-power beamwidths in the E-plane and H-plane as

$$D \approx \frac{4\pi}{\theta_E \theta_H}, \tag{3}$$

where θ_E and θ_H are expressed in radians.

5.4. Antenna Beam Pattern

The beam pattern of an antenna describes the variation of its radiated or received power with direction. It generally consists of a main lobe, which represents the direction of maximum radiation, along with possible side lobes and a back lobe. The beam pattern can be experimentally obtained by measuring the received power as the receiving antenna is rotated through different angles.

5.5. Experimental Setup

The experimental setup consisted of a dipole antenna connected to a signal generator and a horn antenna connected to a spectrum analyzer. For the distance-dependence measurement, the separation d between the transmitting dipole and receiving horn was varied while maintaining their relative alignment. For the beam-pattern measurement, the separation was kept fixed and the horn antenna was rotated about its vertical axis through an angle θ. The received power at each distance or angular position was recorded using the spectrum analyzer.

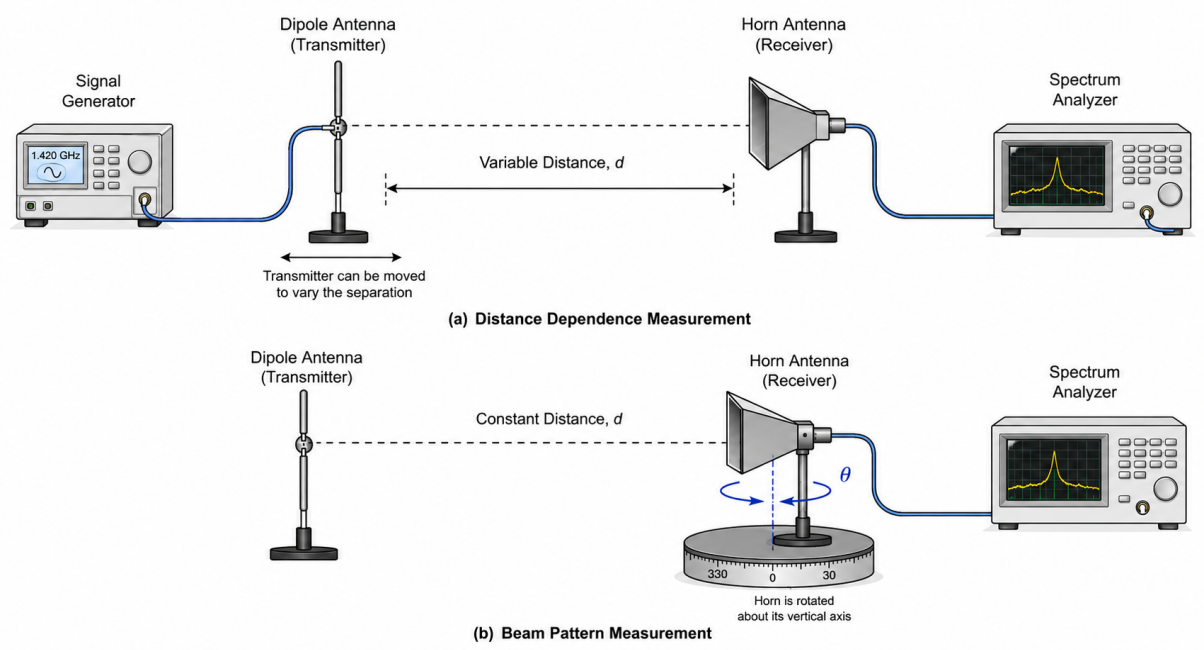


Figure 3: Experimental setup for distance-dependence and beam-pattern measurements.

Table 1: Received power and corresponding $20\log_{10} R$ for different antenna separations.

Distance R (cm)	Received Power P_r (dBm)	$20\log_{10} R$
261	-30.00	48.333
262	-31.50	48.366
292	-29.40	49.308
325	-27.50	50.238
351	-28.40	50.906
383	-30.75	51.664
390	-30.20	51.821
410	-28.30	52.256
417	-27.50	52.403
425	-32.40	52.568
500	-33.55	53.979
674	-36.00	56.573

6. Observations

The experiment was performed under fixed operating conditions for the transmitting and receiving instruments. The signal generator was set to a frequency of 1.4 GHz with an output level of 20. The spectrum analyzer was configured with a reference level of -20 and an attenuation of 20 dB, with the measured signal displayed in dBm. The same frequency and instrument settings were maintained throughout the measurements so that the variation in the recorded power could be attributed primarily to the change in distance or angular orientation.

The measured values of received power for the different experimental configurations are tabulated below. The distance-dependence measurements were obtained by varying the separation between the transmitting dipole and the receiving horn while keeping the instrument settings fixed.

Table 2: Measured received power as a function of angle for the co-polarization configuration.

Angle (°)	Power in E (dBm)	Power in H (dBm)
-180	-49.02	-48.98
-170	-45.02	-49.80
-160	-43.06	-49.23
-150	-42.08	-52.80
-140	-43.11	-53.00
-130	-43.21	-43.63
-120	-42.90	-41.15
-110	-42.51	-40.30
-100	-44.06	-39.87
-90	-44.03	-43.67
-80	-44.12	-50.05
-70	-47.97	-55.20
-60	-48.05	-54.90
-50	-42.97	-47.20
-40	-40.12	-43.63
-30	-38.02	-43.76
-20	-36.64	-38.60
-10	-35.06	-37.42
0	-33.46	-35.07
10	-32.15	-33.62
20	-31.02	-31.02
30	-30.72	-30.72
40	-31.72	-30.16
50	-33.13	-30.56
60	-34.59	-31.47
70	-36.43	-30.99
80	-39.23	-33.72
90	-40.45	-34.98
100	-39.20	-37.33
110	-39.10	-40.06
120	-38.60	-43.46
130	-37.76	-46.28
140	-35.42	-47.04
150	-34.10	-44.02
160	-42.10	-42.63
170	-45.20	-46.59
180	-49.02	-48.98

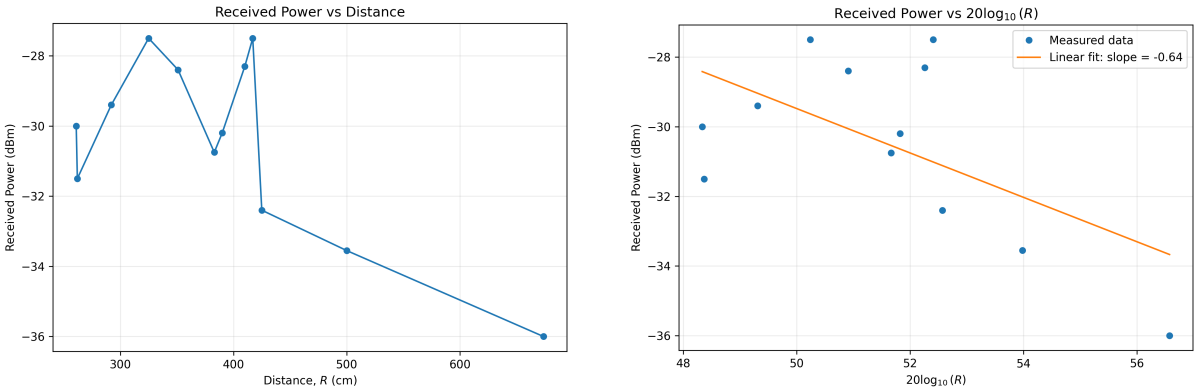
For the beam-pattern measurements, the horn antenna was rotated through different angular positions and the corresponding received power was recorded separately for the E-plane and H-plane configurations. The measurements were performed for both co-polarized and cross-polarized orientations.

The recorded values are presented in Tables 1, 2, and 3.

Table 3: Measured received power as a function of angle for the cross-polarization configuration.

Angle (°)	Power in E (dBm)	Power in H (dBm)
-90	-60.00	-42.27
-80	-55.50	-44.04
-70	-48.40	-46.09
-60	-54.71	-45.24
-50	-52.62	-46.57
-40	-49.17	-49.62
-30	-46.01	-50.07
-20	-48.30	-49.15
-10	-49.40	-43.22
0	-51.50	-43.64
10	-51.31	-41.19
20	-48.20	-41.00
30	-47.30	-40.44
40	-46.80	-39.27
50	-47.20	-42.42
60	-59.60	-43.42
70	-50.50	-47.13
80	-51.80	-48.00
90	-49.50	-48.90

7. Results and Discussion

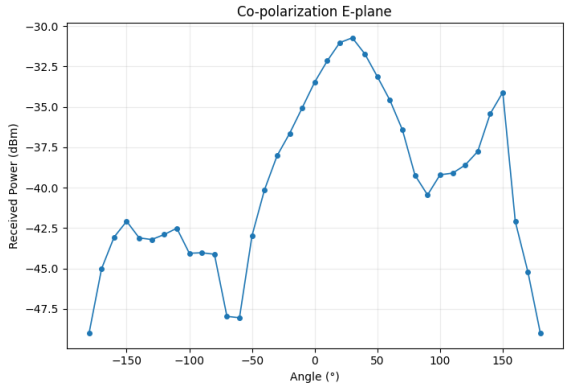


(a) Received power as a function of distance.

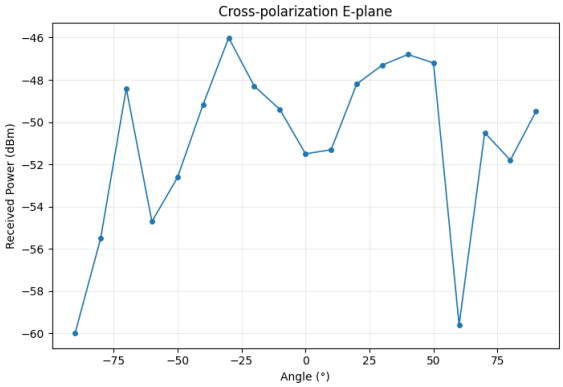
(b) Received power as a function of $20\log_{10}(R)$.

Figure 4: Dependence of received power on the separation between the transmitting and receiving antennas.

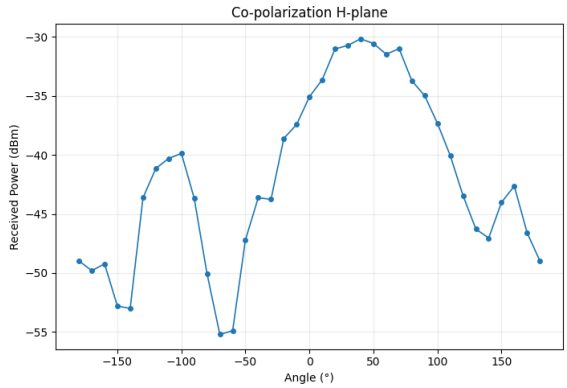
For the distance measurements, as shown in 4 there are noticeable deviations from the expected $1/r^2$ dependence. At relatively large distances, the received power decreases with increasing separation, as expected. However, at shorter distances, the measured power shows significant non-uniformity. This deviation can be attributed the propagation of radio waves within the laboratory environment, where radio waves reflected from the walls and nearby objects interfere with the directly received wave. The resulting constructive and destructive interference causes fluctuations in the measured received power, leading to deviations from the ideal $1/r^2$ behaviour.



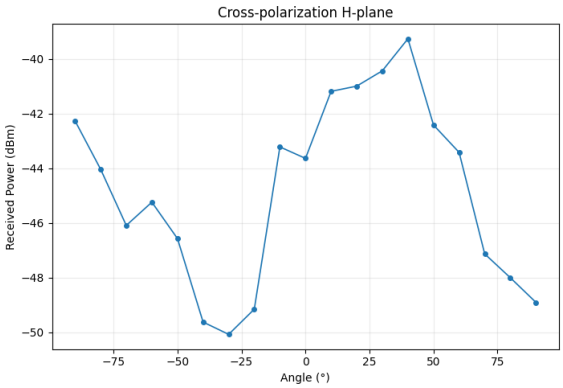
(a) E-plane co-polarization.



(b) E-plane cross-polarization.



(c) H-plane co-polarization.



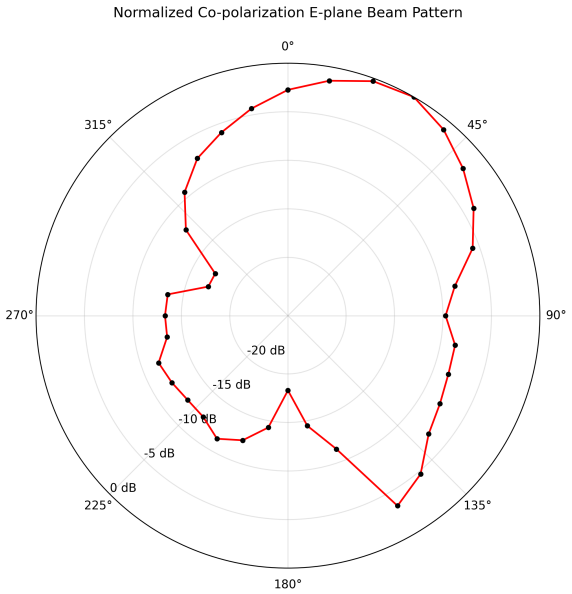
(d) H-plane cross-polarization.

Figure 5: Measured received power as a function of angle for the co-polarization and cross-polarization configurations.

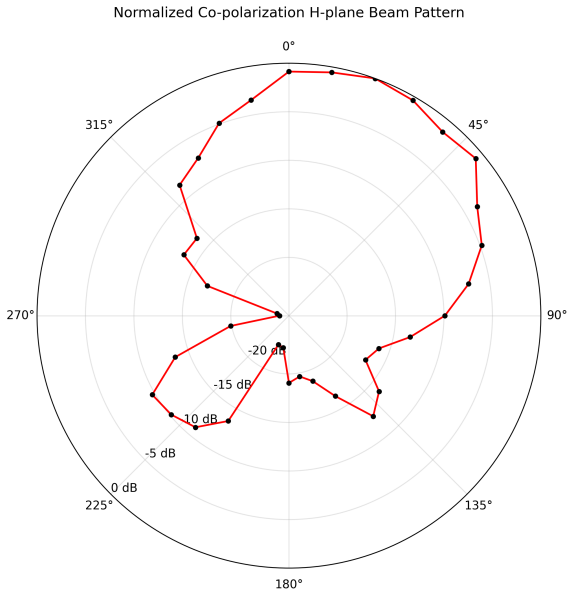
The received power shows a clear dependence on the orientation of the receiving antenna in both the E- and H-planes. In the co-polarization E-plane measurement, the highest received power was approximately -30.72 dBm, while in the H-plane it was approximately -30.16 dBm. The cross-polarization measurements showed lower received powers overall, with the E- and H-plane responses also exhibiting different peak values and angular variations.

Another noticeable feature is the presence of multiple peaks in all four angular measurements. This means that the receiving antenna does not respond only in one particular direction. Instead, significant responses are observed at several angles, indicating a degree of multidirectional behaviour in the measured antenna response. These additional peaks may also be influenced by reflections and interference from the surrounding laboratory environment.

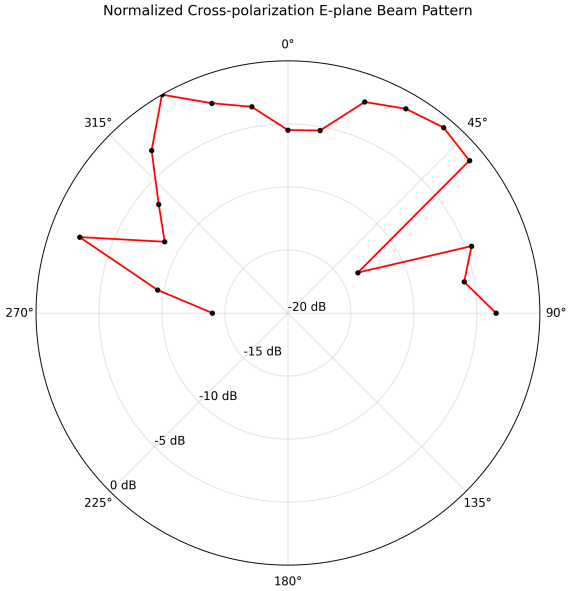
The relative received power is also different between the co-polarization and cross-polarization cases. The co-polarized configuration generally gives a stronger response because the polarization of the incident wave is better aligned with the receiving antenna. The cross-polarized configuration gives a weaker response, although its angular pattern still shows several peaks. A difference is also observed between the E- and H-plane measurements, showing that the antenna response depends not only on the polarization state but also on the plane in which the antenna is rotated. Thus, the measured patterns demonstrate that both antenna orientation and polarization have a significant effect on the received radio-wave power.



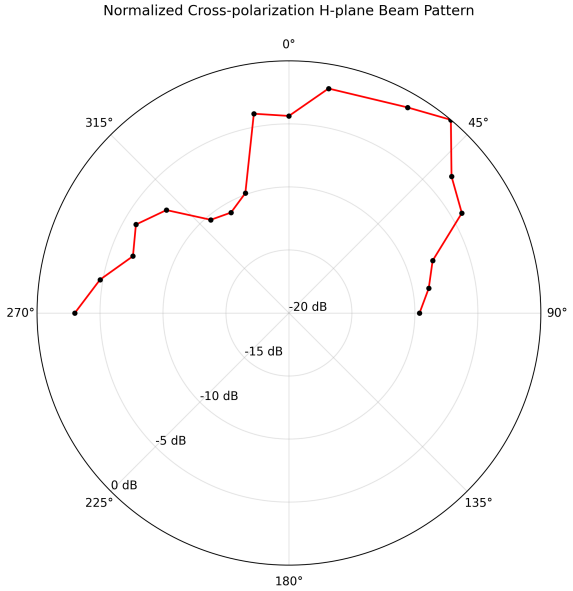
(a) Co-polarization, E-plane.



(b) Co-polarization, H-plane.



(c) Cross-polarization, E-plane.



(d) Cross-polarization, H-plane.

Figure 6: Normalized beam patterns of the receiving antenna for the different polarization and principal-plane configurations.

The normalized beam patterns obtained for the E- and H-planes show that the antenna response is not confined to a single direction. Significant responses are observed over a wide range of angles in all four configurations. This indicates that the antenna exhibits a degree of multidirectionality, rather than behaving as a perfectly directional antenna with a single dominant lobe.

The plots also show the presence of several side lobes and back-lobe responses. In the co-polarization patterns, relatively strong responses are visible away from the main lobe, particularly in the regions around the opposite side of the antenna. Similar secondary features are also present in the cross-polarization patterns. These lobes indicate that the received signal is not completely suppressed when the antenna is oriented away from its main response direction. Some of these additional features may be due to the antenna itself, while reflections from the walls and other objects in the laboratory can also produce constructive and destructive interference and modify the measured pattern.

8. Beamwidth and Directivity Calculation

The maximum received power in the E-plane was measured to be

$$P_{\max,E} = -30.72 \text{ dBm}.$$

The half-power level corresponds to a decrease of approximately 3 dB from the maximum received power. Therefore,

$$P_{-3\text{dB},E} = P_{\max,E} - 3 = -30.72 - 3 = -33.72 \text{ dBm}.$$

The first -3 dB crossing lies between -10° and 0°. Using linear interpolation,

$$\theta_{E,L} = -10 + (0 - (-10))\frac{-33.72 - (-35.06)}{-33.46 - (-35.06)},$$

giving

$$\boxed{\theta_{E,L} \approx -1.63^\circ}.$$

The second -3 dB crossing lies between 50° and 60°. Thus,

$$\theta_{E,R} = 50 + (60 - 50)\frac{-33.72 - (-33.13)}{-34.59 - (-33.13)},$$

giving

$$\boxed{\theta_{E,R} \approx 54.04^\circ}.$$

Hence, the E-plane half-power beamwidth is

$$\text{HPBW}_E = \theta_{E,R} - \theta_{E,L},$$

$$\boxed{\text{HPBW}_E \approx 55.67^\circ}.$$

For the H-plane, the maximum received power was

$$P_{\max,H} = -30.16 \text{ dBm}.$$

Therefore, the corresponding -3 dB level is

$$P_{-3\text{dB},H} = -30.16 - 3 = -33.16 \text{ dBm}.$$

The first -3 dB crossing lies between 10° and 20°. Linear interpolation gives

$$\theta_{H,L} = 10 + (20 - 10)\frac{-33.16 - (-33.62)}{-31.02 - (-33.62)},$$

and hence

$$\boxed{\theta_{H,L} \approx 11.77^\circ}.$$

The second -3 dB crossing lies between 60° and 70°. Thus,

$$\theta_{H,R} = 60 + (70 - 60)\frac{-33.16 - (-31.47)}{-33.72 - (-31.47)},$$

giving

$$\boxed{\theta_{H,R} \approx 67.51^\circ}.$$

Therefore, the H-plane half-power beamwidth is

$$\mathrm{HPBW}_H = \theta_{H,R} - \theta_{H,L},$$

$$\boxed{\mathrm{HPBW}_H \approx 55.74^\circ}.$$

The beamwidths are converted to radians before calculating the directivity:

$$\theta_E = 55.67^\circ \left(\frac{\pi}{180}\right) \approx 0.9716 \text{ rad},$$

$$\theta_H = 55.74^\circ \left(\frac{\pi}{180}\right) \approx 0.9728 \text{ rad}.$$

Using the beamwidth approximation for directivity,

$$D \approx \frac{4\pi}{\theta_E \theta_H},$$

we obtain

$$D \approx \frac{4\pi}{(0.9716)(0.9728)} \approx \boxed{13.25}.$$

The calculated directivity of approximately 13.25 indicates that the antenna has a measurable preference for certain directions, despite the presence of significant side and back lobes in the measured beam patterns.

9. Conclusion

From the experiment, we developed a better understanding of how an antenna works and how one of its important properties, directivity, can be determined from its beam pattern. We also observed that the antenna response is not limited to a single direction, but shows significant responses at multiple angles. The experiment also gave us an indirect understanding of how interactions of radio waves with the surrounding environment, particularly reflections and interference, can affect the measured response of an antenna. Overall, the experiment provided a useful platform to understand the behaviour of radio waves and antennas through direct observation and to draw meaningful conclusions from the experimental measurements.

10. Sources of Error

- Reflections from the walls, tables, and other objects in the laboratory can produce multipath propagation and interference, affecting the measured received power.
- Small errors in the angular positioning and manual rotation of the antenna can introduce uncertainty in the measured beam pattern.
- Imperfect alignment of the transmitting and receiving antennas and their polarization axes can affect the measured co-polarized and cross-polarized responses.
- Instrumental noise and fluctuations in the received signal can cause small variations in the measured power, particularly at lower signal levels.

AI Usage Declaration

The following uses of Artificial Intelligence tools, primarily ChatGPT, were made during the preparation of this report:

1. **Polishing and improving written content:**

   AI was used to improve the clarity, grammar, structure, and overall quality of the written content in the report. The ideas, observations, and experimental interpretations were based on my own understanding and observations, while AI was used to refine their presentation.

2. **Formatting of experimental data into tables:**

   AI was used to assist in converting the experimentally recorded datasets into properly formatted LaTeX tables for inclusion in the report. The original numerical data were provided to the AI directly from the experimental observations, and the resulting tables were checked against the original data.

3. **Generation of plotting and analysis code:**

   AI was used to assist in writing Python code for processing the experimental data and generating the plots presented in this report. This included plots of received power as a function of distance and angle, as well as normalized beam patterns. The experimental data used in these plots were obtained from the measurements performed during the laboratory experiment.

4. **Conceptual understanding of antennas and radio waves:**

   AI was used as a supplementary learning tool to understand the basic functioning of antennas and the physical principles involved in the experiment. This included understanding concepts such as antenna directivity, polarization, beam patterns, and the response of a receiving antenna to an incident electromagnetic field. AI was also used to clarify how the properties and orientation of an antenna, including the dipole, influence the induced electrical response during reception.

5. **Generation of Images used in the theory section :** All the figures that were used in the theory sections (Figs. 1,2,3) were produced by AI based on prompts that were given by me which instructed the AI to generate the required images.

The AI tools were used primarily for writing assistance, data presentation, coding assistance, and conceptual clarification. The experimental measurements, datasets, observations, calculations, and conclusions presented in this report are based on the work performed during the laboratory experiment.

Declaration by Akshar Kishore